A-A-53749A
9 March 1990
SUPERSEDING
A-A-0053749(DM)
13 October 1988

COMMERCIAL ITEM DESCRIPTION

REFRIGERATOR, MECHANICAL, BIOLOGICALS

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a lightweight, portable mechanical refrigerator suitable for use in a field ward.

Salient characteristics:

The unit shall be a lightweight, portable, mechanical refrigerator having a minimum (min.) 6.7 cubic feet (cu. ft.) storage space which includes a 0.7 cu. ft. minimum freezer compartment and drip pan, three shelves, adjustable thermostat, carrying handles, corrosion-resisting steel shield for mounting over rear mechanism opening, and glide feet. The refrigerator temperature range shall be 34°F to 42°F; freezer temperature range shall be 12°F to 20°F.

The refrigerator and modifications shall be in accordance with DPSC Drawing 25044. The refrigerator shall operate satisfactorily from 115 volts, 50/60 Hz. Overall dimensions after modifications shall not exceed a depth of 25 inches, a width of 26 inches and a height of 36 inches. Weight of refrigerator after modification shall not exceed 120 pounds.

AMSC N/A FSC 4110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Outer shell (cabinet) shall be carbon steel approximately 1/32-inch thick, suitably treated and finished in the manufacturer's standard white baking enamel. The mounting plate shall be thoroughly cleaned and treated with a wash primer. Painting shall not be required on the zinc plated carrying handles.

A zinc chromatic primer shall then be applied to a dry film thickness of 0.3 to 0.5 mil. The finish coat shall be a sprayed-on white baking enamel. Coatings shall be applied in accordance with the manufacturer's directions. Inner shell and door liner shall be refrigerator grade ABS plastic or high-impact styrene. The inner shell does not have to be of a one-piece construction, as long as accepted commercial practices are used to fuse the individual components to form the interior and prevent crevices and cavities. Door trim shall be anodized aluminum. The door shall be fitted with magnetic door gaskets for positive seal and magnetic latch. A door securing strap shall be constructed of polyurethane or equal. The securing strap shall be designed to keep the door closed during refrigerator transport. A door activated interior light shall be provided.

The refrigerator shall have a full-width freezer compartment, one sliding plastic shelf, two wire shelves, and one glass shelf, all which run the full width below the freezer compartment.

Refrigeration unit shall be 1/10 HP hermetically sealed and provided with an adjustable thermostat. Shall be supplied with a line power fuse, motor overload protection and a 9 ft. power cord.

Insulation shall be foamed polyurethane or fiberglass and shall be of necessary thickness and efficiency for providing the following pulldown performance.

Ambient Temperature	Pulldown Time Hrs. (max.)	Cycle time (percent)
120°F	3	100
100°F	1-1/4	55
77°F	1/2	34

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Internal compartment temperature shall be 42°F. Electrical power shall be 115 volts, 60 Hz.

Dielectric strength. The unit shall be capable of withstanding without breakdown, for one minute, a potential of 1240 volts, 60 Hz, applied between live and dead metal parts when tested as specified herein.

Identification marking. Each refrigerator shall have legible labels marked with the following:

National Stock Number
Name or registered trademark of manufacturer
Contract or purchase order number
Wiring diagram
Electrical characteristics
Instructions for use (located so as to be readily visible during use)

The label(s) shall be adhesive backed mylar overlays.

Wiring diagram. The diagram shall portray each electrical component by sketch, outline, or symbol; and shall include the name of each component. The wiring diagram shall show the wiring to all components and any coding used for identification of wires.

Underwriters Laboratories Inc. The refrigerator shall conform to UL 471 Standard for Safety - Commercial Refrigerators and Freezers. Acceptable evidence of this shall be a UL listing mark or a certified test report from an independent laboratory acceptable to the Government.

Workmanship. The refrigerator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One refrigerator, as specified, constitutes one unit.

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Operators' and Service Data Manuals.

The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures will be written in the English language and all schematics and wiring diagrams will use American electrical and electronics symbols. The manuals will include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 4 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one refrigerator, as specified in this description.

Sampling for tests. Sampling for tests including dimensional test, shall be conducted in accordance with MIL-STD, with an AQL of 2.5 (percent defective) and an inspection level of S-1. The unit of product for test purposes shall be one refrigerator.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Operation test. The unit shall be operated for one hour according to manufacturer's directions to verify that the controls are adjusted and function. The unit shall be inspected for damage or leakage from any part, piping or fitting.

Dielectric strength test. A slowly rising potential, from 0 to 1240 volts, 60 Hz, shall be applied between live and dead metal parts. The potential shall be maintained at maximum for one minute.

Pulldown performance test. The unit shall obtain an internal compartment temperature of 42°F with the ambient temperatures maintained as specified herein, within the times specified, and without exceeding the percentage running times specified for at least 24 hours of steady state operation under the conditions for ambient temperature, pulldown time and cycle time specified herein.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

4110-00-115-6027

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 4110-0435

Location: CID/REFRIG/MECH/BIO/H1B-25

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